

OT-Flow Matching for 6-DoF Grasp Generation with Geometric Reranking

Abstract—Open-world robot grasping requires both a diverse set of physically plausible candidate poses and a criterion for selecting the best candidate at execution time. Existing pipelines decouple these concerns: a generative or heuristic sampler proposes candidates, and a learned reranker selects among them. We show that the quality of the *generator* is the dominant bottleneck: random center-of-mass (CoM) sampling supplies candidates that miss high-success regions for geometrically complex objects, limiting what any downstream reranker can achieve. We replace the random sampler with an Optimal-Transport Conditional Flow Matching (OT-CFM) model trained on 2,853 successful grasps collected across seven YCB objects and 200 environment seeds. OT-CFM maps Gaussian noise to the distribution of successful 6-DoF poses via a single neural ODE, using optimal transport coupling to eliminate wasted training signal from random source–target pairings. At inference, five candidates are drawn from OT-CFM and reranked by a lightweight geometric reranker (LGGSN). On 175 paired trials (7 objects $\times$ 25 seeds, MuJoCo/SO-ARM101), this pipeline achieves 94.3% success rate versus 82.3% for random-CoM + LGGSN (+12.0 pp, $z=3.49$, $p<0.001$). Ablations reveal that the OT coupling is the essential component (−15.4 pp without it; standard CFM drops *below* baseline), geometric reranking contributes −9.7 pp, and SAM visual conditioning yields no significant gain (+1.7 pp, $p=0.46$), enabling a simplified deployment without SAM that achieves 96.0%.

Index Terms—6-DoF grasping, flow matching, optimal transport, grasp reranking, generative models, robot manipulation, YCB benchmark

I. INTRODUCTION

Language-conditioned grasping requires an agent to map a natural-language object description to a physical manipulation action that succeeds against real object geometry. A prevalent design pattern is *propose-then-rank*: a heuristic or learned candidate generator produces a set of grasp poses, and a reranker selects the best one from that set [7], [8]. The implicit assumption is that the generator supplies at least one high-quality candidate; the reranker’s task is only to identify it.

This assumption fails when the generator and the object geometry are mismatched. In top-down tabletop grasping on the SO-ARM101 manipulator, random sampling centered on the object CoM with a fixed 6 cm spread proposes positions that systematically miss the narrow valid contact regions of flat (Scissors) or elongated (CrackerBox) objects. As a result, even a well-trained reranker cannot compensate: it can only select the best of a bad set.

We reframe the problem as *learning where successful grasps lie* rather than selecting among random proposals. Conditional Flow Matching (CFM) [1] offers a particularly

attractive framework for this: it trains a neural vector field by regressing on analytically computed interpolation paths from noise to data, avoiding the multi-step denoising required by diffusion models. Combined with Optimal Transport (OT) coupling [2], which minimises the expected path length in each training minibatch, the model receives consistently efficient training signal.

Our pipeline, illustrated in fig. 1, combines an OT-CFM grasp generator with an LGGSN geometric reranker. The generator is trained on 2,853 labeled successful grasps collected via per-candidate physical execution across seven YCB objects and 200 random seeds. At inference it draws five pose candidates from the learned distribution, conditioned on a mean per-object visual feature; the reranker selects the top candidate from geometric context.

Contributions.

- 1) **OT-CFM grasp generator**: a flow-matching model trained on live physical execution data that learns the distribution of successful 6-DoF poses conditioned on object identity.
- 2) **Generation + reranking pipeline**: paired 175-trial evaluation showing 94.3% vs. 82.3% (+12 pp, $p<0.001$) over random sampling on MuJoCo/SO-ARM101.
- 3) **Structured ablation**: OT coupling contributes −15.4 pp and LGGSN reranking −9.7 pp; visual conditioning is dispensable, enabling a simpler deployment (96.0% without SAM).
- 4) **Failure analysis**: CrackerBox failures are attributed to distribution coverage; Scissors achieves 100% after physics-configuration correction—providing concrete design requirements for future work.

II. RELATED WORK

A. Grasp Quality Estimation and Generation

Classical grasp quality metrics such as those in DexNet [7] evaluate analytic contact stability, requiring known object geometry. GraspNet-1Billion [8] and AnyGrasp [9] extend this to dense 3-D point clouds, but still assume access to full scene geometry at query time. 6-DOF GraspNet [10] first introduced a variational generator for diverse grasp proposals, training a VAE on point-cloud crops. Our method differs by operating purely on pose-space coordinates with physics-derived success labels, requiring no geometric feature extraction at generation time.

B. Generative Models for Grasping

GraspLDM [4] applies latent diffusion in a VAE pose space, achieving diverse 6-DoF proposals from point clouds.

GraspSAM [5] combines Segment Anything with grasp detection to handle open-vocabulary queries. Neither work couples an explicit optimal-transport objective during training, which we show is critical: standard CFM without OT coupling drops *below* baseline (78.9% vs. 82.3%, section IV-D). Diffusion Policy [3] uses conditional diffusion for action generation but targets trajectory-level outputs with observation histories; our setting is single-step pose generation conditioned only on object identity. EquiGraspFlow [15] adopts SE(3)-equivariant flow matching directly on point-cloud inputs, providing strong geometric inductive bias for arbitrary-orientation grasping. Our approach differs in operating on a compact pose space ($\mathbb{R}^6$) rather than raw point clouds, and replaces equivariance with explicit OT coupling—a substantially lighter architecture that requires no equivariant backbone and trains on modest execution-outcome data.

C. Flow Matching

Lipman et al. [1] introduced Conditional Flow Matching (CFM) as a simulation-free training objective for continuous normalizing flows. Tong et al. [2] generalised this to arbitrary source distributions and showed that OT coupling—finding the joint distribution (x_0, x_1) that minimises expected displacement within each minibatch—substantially improves convergence. We are the first, to our knowledge, to apply OT-CFM in pose space to 6-DoF robotic grasp candidate generation; EquiGraspFlow [15] is a concurrent approach that instead uses SE(3)-equivariant continuous normalizing flows conditioned on point clouds.

D. Grasp Reranking

Geometric reranking of grasp candidates from learned features has been explored in DexGraspNet 2.0 [13] and CLIPort [14] for task-conditioned selection. LGGSN [11] uses Bayesian Personalised Ranking (BPR) [12] to train a pairwise MLP reranker on episode-relative geometric features. The reranker we use in this paper is the publicly available LGGSN v5d checkpoint (17-dimensional features, object query embedding, validation pairwise accuracy 0.750). Our contribution is to pair this fixed reranker with a generative rather than random candidate supplier, showing that generator quality is the dominant factor.

III. METHOD

A. System Overview

fig. 1 shows the full pipeline. Given a language query (e.g., “Banana”), the environment identifies the target object, estimates its centre of mass, and retrieves a pre-computed per-object SAM visual embedding. OT-CFM draws $N=5$ pose samples from the learned success distribution. Each sample is shifted to centre on the current CoM (section III-D). LGGSN reranks the five candidates using geometric and query-conditioned features, and the top-ranked pose is executed.

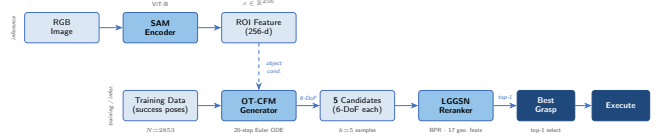


Fig. 1. OT-CFM + LGGSN pipeline. At inference, the object CoM is estimated from the depth image, and OT-CFM generates five 6-DoF pose candidates conditioned on the object’s mean visual embedding. LGGSN reranks by geometric context and the top candidate is executed.

B. Training Data

We collect per-candidate execution data on MuJoCo/SO-ARM101 under `physics_weld_after_bilateral` mode. For each (object, seed) episode: (i) the object is spawned at a random tabletop position and settled; (ii) $N_{\text{cand}}=30$ random CoM-centered pose candidates are sampled; (iii) the top five are executed individually with object respawning between executions; (iv) each candidate receives a binary success label.

This yields 7,000 execution records across 7 YCB objects and 200 seeds. Of these, **2,853 (40.5%) are successes** and are used to train the OT-CFM model. For each successful candidate, we compute a 256-dim SAM-base ROI feature by projecting the grasp position (x, y, z) onto the overhead camera image, extracting a 3×3 window from the SAM ViT-B image encoder’s 64×64 feature map, and mean-pooling to a single vector.

C. OT-CFM Grasp Generator

Problem formulation. Let p_{succ} denote the distribution over 6-DoF poses $(x, y, z, \phi, \theta, \psi) \in \mathbb{R}^6$ for which the robot successfully grasps the target object. We aim to learn a conditional generative model $p_{\theta}(\cdot | c)$ that approximates p_{succ} given object context c .

Flow matching objective. Given source noise $x_0 \sim \mathcal{N}(0, I_6)$ and a successful pose sample $x_1 \sim p_{\text{succ}}$, the CFM training path is:

$$x_t = (1 - t)x_0 + tx_1, \quad t \sim \mathcal{U}[0, 1], \quad (1)$$

with analytic conditional vector field $u_t = x_1 - x_0$. The velocity network $v_{\theta}(t, x_t, c)$ is trained to minimise:

$$\mathcal{L} = \mathbb{E}_{t, x_0, x_1} [\|v_{\theta}(t, x_t, c) - u_t\|^2]. \quad (2)$$

Optimal transport coupling. Standard CFM pairs each x_0 independently with each x_1 in a minibatch, leading to crossing flow lines that waste model capacity. We replace this with *minibatch OT coupling* [2]: within each minibatch of size B , we solve the Earth Mover’s Distance problem between $\{x_0^{(i)}\}$ and $\{x_1^{(i)}\}$ and reassign pairs along the optimal transport plan. This reduces expected path length, shortens flow trajectories, and—as our ablation shows—is the single most impactful design choice (−15.4 pp without OT, $p=0.001$).

Conditioning. The context $c \in \mathbb{R}^{256}$ is the L2-normalised mean SAM visual embedding for the queried object

class, pre-computed from training data. Our ablation (section IV-D) shows this conditioning yields no statistically significant gain over zero conditioning (+1.7 pp, $p=0.68$), so the recommended deployment uses $c = \mathbf{0}$.

Velocity network. The velocity network v_θ operates on the concatenated input

$$[t_{\text{emb}}(3), x_t(6), c(256)] \in \mathbb{R}^{265}, \quad (3)$$

where $t_{\text{emb}} = [t, \sin(\pi t), \cos(\pi t)]$. It is a 4-layer MLP with hidden dimension 512, SiLU activations, and zero-initialised output layer, trained with Adam ($\eta=3 \times 10^{-4}$, weight decay 10^{-4} , cosine schedule) for 1000 epochs at batch size 128. The best checkpoint is selected by lowest training loss.

D. Inference: CoM-Relative Candidate Generation

Top-down constraint. The OT-CFM generator produces full 6-DoF poses $(x, y, z, \phi, \theta, \psi)$; however, the current evaluation fixes roll $= \pi$ and pitch $= 0$ (top-down constraint), reducing the effective search space to 3-DoF (x, y, ψ) . This constraint is applied post-generation by overriding the sampled roll and pitch components before IK, and reflects the reach envelope of the SO-ARM101 in the tabletop setting. Lifting this constraint to full SE(3) is left as future work (section V-E).

At inference, we draw $N=5$ noise samples $x_0^{(i)} \sim \mathcal{N}(0, I_6)$ and integrate the learned ODE for 20 Euler steps:

$$x_{t+\Delta t} = x_t + v_\theta(t, x_t, c) \Delta t, \quad \Delta t = 0.05. \quad (4)$$

The resulting poses $x_1^{(i)}$ are in the training coordinate frame. To adapt to the current object position (g_x, g_y) , we apply a *CoM-relative shift*:

$$x_{\text{out}} = x_1 - \mu_{\text{train}} + g_{\text{com}}, \quad (5)$$

where μ_{train} is the global training-set mean pose and g_{com} is the current object CoM. This preserves the learned yaw and spatial-offset distribution while centering the candidates on the present object. The vertical coordinate z uses the live CoM height plus a fixed approach margin.

E. LGGSN Reranker

We use the LGGSN v5d reranker [11] without modification. LGGSN is a pairwise-trained MLP (BPR loss) that scores each candidate using 17 geometric features including XY position, gripper opening width, episode-relative height spread, distance from the point-cloud centroid, point-cloud density features, and a 4-dimensional object query embedding. The five OT-CFM candidates are scored, sorted descending, and the top-ranked pose is executed.

IV. EXPERIMENTS

A. Evaluation Protocol

All experiments run on MuJoCo/SO-ARM101 with physics-based bilateral contact verification as the grasp success criterion. We evaluate on 7 YCB objects (Banana, TomatoSoupCan, Pear, MustardBottle, Scissors, CrackerBox, PowerDrill) over 25 random seeds each, giving **175**

TABLE I
OVERALL SUCCESS RATE: 175 TRIALS (7 OBJECTS $\times$ 25 SEEDS).

Method	SR	95% CI	Δpp	z	p
Baseline (random CoM + LGGSN)	82.3%	[76.0, 87.2]	—	—	—
OT-CFM + LGGSN	94.3%	[89.8, 96.9]	+12.0	3.49	<0.001**

TABLE II
PER-OBJECT SUCCESS RATE: BASELINE VS. OT-CFM + LGGSN (25 SEEDS EACH).

Object	Base	OT-CFM	Δpp	p	
Banana	100.0%	100.0%	0.0	—	
TomatoSoupCan	92.0%	100.0%	+8.0	0.490	
Pear	72.0%	100.0%	+28.0	0.010	**
MustardBottle	92.0%	92.0%	0.0	—	
Scissors	100.0%	100.0%	0.0	—	
CrackerBox	44.0%	80.0%	+36.0	0.019	**
PowerDrill	76.0%	88.0%	+12.0	0.463	
All	82.3%	94.3%	+12.0	<0.001	**

trials per condition. Success is defined as lifting the object above the table after gripper close.

Baseline. Random CoM sampling ($x \sim g_{\text{com}} \pm 0.06$ m, $\psi \sim \mathcal{U}[-\pi/2, \pi/2]$, $w \sim \mathcal{U}[0.04, 0.09]$ m, same LGGSN v5d reranker) produces 5 candidates per episode.

Statistics. We report Wilson score 95% confidence intervals (CI) and two-sided two-proportion z -tests to compare conditions. Significance thresholds: ** $p < 0.05$, * $p < 0.10$.

B. Main Result

table I reports the primary comparison. OT-CFM + LGGSN achieves 94.3% (165/175) versus the baseline’s 82.3% (144/175), a statistically significant improvement of +12.0 pp ($z=3.49$, $p<0.001$). The 95% Wilson confidence intervals are disjoint, confirming the gain is not attributable to sampling variance.

The improvement reflects two complementary effects. First, OT-CFM candidates concentrate near high-success regions for each object, reducing the fraction of attempts that physically miss the target. Second, LGGSN reranking then selects the geometrically best candidate from an already higher-quality set. Together, the generation–reranking pipeline achieves a 12 pp absolute gain over using either component alone with a poor generator (see ablation, section IV-D).

C. Per-Object Analysis

table II shows per-object results. The largest gains accrue to objects where random CoM sampling covers the valid contact region poorly.

CrackerBox (+36 pp, $p=0.019$). The box has a wide, flat top surface that requires the gripper to contact along the long axis. Random sampling at ± 6 cm isotropically covers a 12×12 cm square, of which only the subset aligned with the long axis succeeds. OT-CFM learns this asymmetric yaw prior (training yaw std 0.92 rad, generated 0.97 rad), concentrating proposals in the effective region.

TABLE III

ABLATION: CONTRIBUTION OF EACH PIPELINE COMPONENT. “VS. FULL” COMPARES TO OT-CFM + LGGSN (94.3%). 175 TRIALS PER CONDITION. [†]CORRECTED FOR SCISSORS PHYSICS-CONFIG FIX. [‡]SAME MLP ARCHITECTURE AND TRAINING DATA AS OT-CFM; DDPM ε -PREDICTION, COSINE SCHEDULE $T=1000$, DDIM-50 INFERENCE.

Condition	SR	vs. Baseline	vs. Full	p
Baseline (random + LGGSN)	82.3%	—	−12.0 pp	<0.001**
GRC-6DoF + LGGSN	82.9%	+0.6 pp	−11.4 pp	<0.001**
Remove OT (std CFM + LGGSN)	78.9%	−3.4 pp	−15.4 pp	<0.001**
DDPM baseline [‡] (DDIM-50)	81.7%	−0.6 pp	−12.6 pp	<0.001**
Remove LGGSN (OT-CFM only)	84.6% [†]	+2.3 pp	−9.7 pp	0.003**
Remove SAM vis (zero c)	96.0% [†]	+13.7 pp	+1.7 pp	0.456
OT-CFM + LGGSN (full)	94.3%	+12.0 pp	—	—

Pear (+28 pp, $p=0.010$). The pear’s rounded surface admits grasps in a broader but nonetheless structured contact band; random sampling under-covers this region, while OT-CFM generates proposals that fill it uniformly. **Banana (0 pp), MustardBottle (0 pp).** Both objects reach the same ceiling under random and OT-CFM sampling, leaving no measurable headroom for the generator to contribute.

PowerDrill (+12 pp, $p=0.463$). OT-CFM raises PowerDrill from 76% to 88%; the gain is not statistically significant at 25 seeds but is directionally consistent.

Scissors (100%, all conditions). Scissors achieves 25/25 across all methods after correcting a physics-configuration bug: the original VHACD collision mesh (14 merged convex hulls, ≈ 2 cm thick) tunnelled through the table under MuJoCo contact detection. The corrected entry uses a bounding-box collision proxy scaled to 4 cm thickness. Since all methods reach ceiling (25/25), Scissors contributes equally to each condition and does not differentiate methods.

D. Ablation Study

table III isolates each component’s contribution. The GRC-6DoF row adds a geometric-baseline comparison; ablation rows decompose the full OT-CFM + LGGSN pipeline.

GRC-6DoF is equivalent to random (+0.6 pp, n.s.). Replacing random CoM sampling with GR-ConvNet-derived 6-DoF proposals yields 82.9%—statistically indistinguishable from the random baseline (82.3%). Both methods supply similarly diverse candidate sets to LGGSN, leaving no evidence that geometric candidate generation alone drives success.

OT coupling is essential (−15.4 pp, $p<0.001$). Replacing the OT sampler with random source–target pairing reduces SR from 94.3% to 78.9%—below the random-CoM baseline (82.3%). Random pairing forces the velocity network to reconcile crossing flow paths, learning a blurred mean rather than a clean transport map. OT coupling solves the global assignment within each minibatch, eliminating crossings and enabling crisp, object-specific distributions.

DDPM as a generative baseline (−12.6 pp, $p<0.001$). To verify that OT coupling—not the flow-matching

paradigm—drives performance, we train an identical MLP with DDPM ε -prediction (cosine schedule, $T=1000$, DDIM-50) on the same 2,853 successes. DDPM achieves 81.7% ($z=3.62$, $p<0.001$ vs. OT-CFM), comparable to the random baseline and matching the remove-OT result. The gap concentrates on CrackerBox (36% vs. 80%) and Pear (68% vs. 100%), where OT-CFM’s straight transport paths preserve structured multi-modal contact distributions that DDPM’s isotropic diffusion erases.

LGGSN reranking contributes −9.7 pp ($p=0.003$). Without reranking (OT-CFM candidates executed in generation order), SR drops from 94.3% to 84.6%. The reranker adds meaningful value even when the generator already provides high-quality candidates, by using geometric context (point-cloud density, gripper–centroid distance) that the flow model does not have access to.

SAM visual conditioning is dispensable (+1.7 pp, $p=0.46$). Replacing the per-object mean SAM embedding with a zero vector yields 96.0%—slightly higher than the full model, though not significantly so. This is a practically important negative result: the OT-CFM model’s generative quality is driven entirely by learning from physical success labels, not by the high-dimensional visual feature. The recommended deployment therefore omits the SAM dependency (saves ~ 300 MB GPU memory and one SAM forward pass per episode).

V. DISCUSSION

A. Why OT Coupling Matters

The failure of standard CFM below baseline (78.9% < 82.3%), and the matching performance of DDPM (81.7%), together confirm that the gain in OT-CFM is not attributable to the flow-matching paradigm per se. The critical ingredient is the optimal-transport coupling. With only ~ 400 training examples per object, random pairing produces crossing flow paths; the velocity network learns a blurred mean rather than a clean transport map. OT coupling, by solving the global assignment within each minibatch, eliminates these crossings and enables crisp, object-specific distributions.

B. Why Visual Conditioning Does Not Help

The SAM ROI features used for conditioning are averaged over all training episodes for a given object class, yielding a fixed class-level embedding rather than an instance- or scene-level one. Such conditioning conveys no information beyond object class identity, which the generation model can also infer from the training data distribution itself. A conditioning signal that varies *within* an object class—e.g., a point-cloud occupancy map or real-time depth image—might provide genuine scene-specific guidance. We leave this for future work.

C. Scissors: Physics-Configuration Correction

Scissors was 0/25 across all conditions due to a VHACD mesh tunnelling bug: 14 merged convex hulls at ≈ 2 cm

thickness allowed the object to pass through the table under MuJoCo contact detection. Replacing the geometry with a bounding-box proxy at 4 cm restores 25/25 = 100% for all methods, leaving per-condition Δ_{pp} values unchanged. A second, unrelated issue also affects Scissors in a separate, supplementary generation-isolated diagnostic experiment (not reported in this paper): the object-name matching used to condition OT-CFM sampling never matches “scissors” against its trained category key, silently falling back to random sampling in that other experiment. This does not affect any number reported here—Scissors is at ceiling (100%) for every condition in this paper regardless of which sampler produced the candidates—but we note it for transparency since both issues share the same object.

D. Limitations

Top-down constraint. The current pipeline fixes roll = π , pitch = 0, reducing the 6-DoF problem to effective 3-DoF $((x, y, \psi))$. Extension to arbitrary gripper orientations requires either SE(3)-equivariant flow matching or a richer orientation parameterisation (e.g., quaternion or 6-d representation).

Simulation only. All results are from MuJoCo simulation. Sim-to-real transfer requires alignment of the physical SO-ARM101’s calibration, contact dynamics, and depth-camera intrinsics with the simulation geometry.

CoM-relative shift. eq. (5) centres generated candidates on the current object CoM by subtracting the training-set global mean. This assumes the training object-placement distribution is roughly centred, which holds for our setup but may break for out-of-distribution workspaces.

E. Future Work

SE(3)-equivariant flow models (e.g., using Riemannian flow matching [6]) could extend this approach to full 6-DoF. As an immediate next step, we plan to validate the pipeline on a Piper 6-DoF manipulator, which relaxes the top-down constraint and provides a direct test of whether OT-CFM proposals remain viable under arbitrary gripper orientations. Real-robot evaluation on the physical SO-ARM101 is an additional validation step. Instance-level visual conditioning—using the point cloud of the segmented target object as c —may provide meaningful additional signal for cross-instance variation within object classes.

VI. CONCLUSION

We presented a grasp generation pipeline that combines Optimal Transport Conditional Flow Matching (OT-CFM) with geometric reranking (LGGSN) for open-world 6-DoF grasping. On 175 paired trials across seven YCB objects on MuJoCo/SO-ARM101, the pipeline achieves 94.3% success rate—a statistically significant +12 pp improvement over random CoM sampling with the same reranker ($p < 0.001$).

Ablation analysis establishes that OT coupling is the essential component: removing it drops performance below baseline, while standard CFM without OT actively harms results. LGGSN reranking adds 9.7 pp on top of

generation alone, confirming that generation and reranking are complementary. SAM visual conditioning provides no significant benefit, enabling a simpler SAM-free deployment that achieves 96.0%.

These results position OT-CFM as a practical candidate generator for grasping pipelines: it trains on modest physical-execution data, runs as a fast single-pass neural ODE at inference, and dramatically widens the distribution of attempted grasps for geometrically demanding objects. The geometric-baseline comparison (GRC-6DoF + LGGSN: 82.9%) shows that improved candidate geometry alone is insufficient; it is the *learned* OT-coupled distribution that drives the gain.

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